

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX)		**Date:01-01-2025**
Time: 3HRS	**GTM-8**	**Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. The co-ordinates of a moving particle at a time t, are given by, $x = 5\sin 10t, y = 5\cos 10t$. Considering all quantities in SI units, the speed of the particle is ____

A) 25 B) 50 C) 10 D) None

2. A wire has a mass (0.3 ± 0.003)g, radius (0.5 ± 0.005) mm and length (6 ± 0.06)cm. The maximum percentage error in the measurement of its density is

A) 1 B) 2 C) 3 D) 4

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3. Two particles are projected from the same point on ground simultaneously with speed 20 m/s and $\frac{20}{\sqrt{3}} m/s$ at angles 30° and 60° respectively with the horizontal in the same direction. The maximum distance between them till both of them strike the ground is approximately ($g = 10\ m/s^2$)

A) 23.1 m B) 16.4 m C) 30.2 m D) 10.4 m

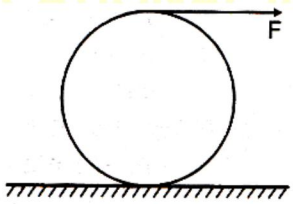
4. A block of mass 0.1 kg is held against a wall by applying a horizontal force of 5 N on the block. If the coefficient of friction between the block and the wall is 0.5, the magnitude of the frictional force acting on the block is

A) 2.5 N B) 0.98 N C) 4.9 N D) 0.49 N

5. An ideal spring with spring-constant k is hung from the ceiling and a block of mass M is attached to its lower end. The mass is released with the spring initially unstretched. Then the maximum extension in the spring is

A) $\frac{4Mg}{k}$ B) $\frac{2Mg}{k}$ C) $\frac{Mg}{k}$ D) $\frac{Mg}{2k}$

6. A force F is applied at the top of a ring of mass M and radius R placed on a rough horizontal surface as shown in figure. Friction is sufficient to prevent slipping. The friction force acting on the ring is

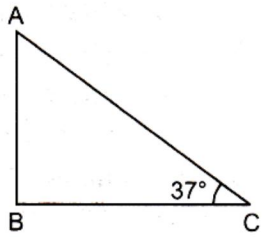


A) $\frac{F}{2}$ towards right B) $\frac{F}{3}$ towards left

C) $\frac{2F}{3}$ towards right D) zero

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7. ABC is a right angled triangular plate of uniform thickness. I_1, I_2 and I_3 are moments of inertia about AB, BC and AC respectively. Then which of the following relation is correct ?



A) $I_1 = I_2 = I_3$ B) $I_2 > I_1 > I_3$ C) $I_3 < I_2 < I_1$ D) $I_3 > I_1 > I_2$

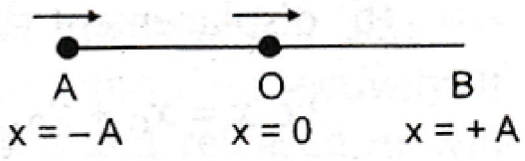
8. Two satellite A and B, having masses in the ratio of 4:3 , are revolving in circular orbit of radii 3r and 4r respectively around the earth the ratio of total mechanical energy of A to B is

A) 9:16 B) 16:9 C) 1:1 D) 4:3

9. A satellite is revolving round the earth with orbital speed v_0. If satellite is stopped suddenly and released, the speed with which it will strike the surface of earth would be (v_e = escape velocity of a particle on earth's surface)

A) $\frac{v_e^2}{v_0}$ B) v_0 C) $\sqrt{v_e^2 - v_0^2}$ D) $\sqrt{v_e^2 - 2v_0^2}$

10. Two particles undergo SHM along the same line with the same time period (T) and equal amplitudes (A). At a particular instant one particle is at x = - A and the other is at x = 0. They move in the same direction just after this moment. They will cross each other at



A) $x = A$ B) $x = \frac{\sqrt{3}A}{2}$ C) $x = \frac{A}{2}$ D) $x = \frac{A}{\sqrt{2}}$

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11. A balloon with mass M is descending down with an acceleration a (a < g). What mass of its contents must be removed so that it starts moving up with same acceleration a ?

A) $\frac{2M(g)}{a}$ B) $\frac{M(g+a)}{g}$ C) $\frac{Mg}{g+a}$ D) $\frac{2Ma}{g+a}$

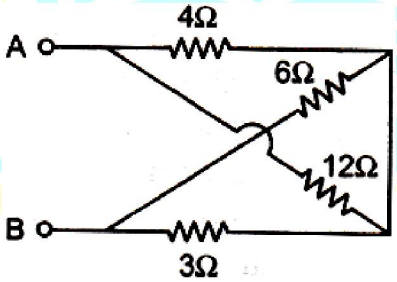
12. An open pipe is suddenly closed at one end with the result that the frequency of third harmonic of the closed pipe is found to be higher by 100 Hz than the fundamental frequency of the open pipe. The fundamental frequency of the open pipe is

A) 200 Hz B) 300 Hz C) 240 Hz D) 480 Hz

13. A particle of charge q and mass m starts moving from the origin under the action of an electric field $\vec{E} = E\hat{i}$ and magnetic field $\vec{B} = B\hat{i}$ with a velocity $\vec{v} = v_0\hat{j}$. The speed of the particle will become $2v_0$ after a time

A) $t = \frac{2mv_0}{qE}$ B) $t = \frac{2Bq}{mv_0}$ C) $t = \frac{\sqrt{3}Bq}{mv_0}$ D) $t = \frac{\sqrt{3}mv_0}{qE}$

14. In the given network, the equivalent resistance between A and B is



A) 6Ω B) 16Ω C) 7Ω D) 5Ω

15. In ideal Young's double slit experiment with λ as wavelength and d as slit separation, intensity at a point is (1/4) of the maximum intensity. Angular position of such a point closest to central maximum is:

A) $\sin^{-1}\left(\frac{\lambda}{d}\right)$ B) $\sin^{-1}\left(\frac{\lambda}{2d}\right)$ C) $\sin^{-1}\left(\frac{\lambda}{3d}\right)$ D) $\sin^{-1}\left(\frac{\lambda}{4d}\right)$

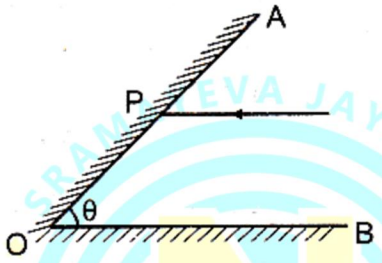
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16. Two point sources S_1 and S_2 are 24 cm apart. Where should a convex lens of focal length 9 cm be placed in between them so that the images of both sources are formed at the same place ?

A) 6 cm from S_1 B) 15 cm from S_1 C) 10 cm from S_1 D) 12 cm from S_1

17. Two plane mirrors are inclined at angle θ as shown in figure. If a ray parallel to OB strikes the other mirror at P and finally emerges parallel to OA after undergoing a total of two reflections then θ is equal to



A) 90° B) 60° C) 45° D) 30°

18. One calorie is defined as the amount of heat required to raise temperature of 1 g of water by 1°C and it is defined under which of the following conditions ?

A) From 14.5°C to 15.5°C at 760 mm of Hg

B) From 98.5°C to 99.5°C at 760 mm of Hg

C) From 13.5°C to 14.5°C at 76 mm of Hg

D) From 3.5°C to 4.5°C at 76 mm of Hg

19. Maximum speed of a particle in simple harmonic motion is v_{max}. Then average speed of a particle in SHM for complete cycle is equal to

A) $\frac{v_{max}}{2}$ B) $\frac{v_{max}}{\pi}$ C) $\frac{\pi v_{max}}{2}$ D) $\frac{2v_{max}}{\pi}$

20. A girl throws a ball with initial velocity v at an inclination of 45°. The ball strikes a smooth vertical wall at a horizontal distance d from the girl and after rebounding return to her hand. What is the coefficient of restitution between the wall and the ball?

A) $\frac{v^2 - gd}{gd}$ B) $\frac{gd}{v^2 - gd}$ C) $\frac{gd}{v^2}$ D) $\frac{v^2}{gd}$

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. Capillaries of length ℓ and 2ℓ are connected in series. Their radii are r and 2r respectively. If stream line but viscous flow is maintained and pressure difference across first and second capillaries are P_1 and P_2 respectively. Then the ratio $\frac{P_1}{P_2}$ will be

22. If the period of rotation of electron in hydrogen first orbit is T then, the period of rotation of electron in He^+, second orbit is equal to $n \times T$ then the value of n ?

23. The ratio of RMS output voltage of full wave rectifier and half wave rectifier is $\sqrt{n}$. The value of n is ? (The input voltage is sine wave).

24. The two lake, one having surface area A and another having surface area 2A, are exposed to negative temperature in Celsius scale. The time taken by first lake in freezing 1 cm is T then the time taken by the second lake to freeze by 1 cm is $\frac{T}{n}$. The value of n is ?

25. The four different liquids having specific heat capacity are in the order of s, 2s, 3s and 4s and their respective temperatures are in the order of 100°C, 80°C, 20°C and 10°C. If they are mixed having the same masses, the final temperature of the mixture in $°C$ is ?

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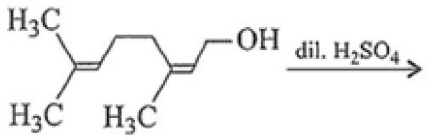
CHEMISTRY **MAX.MARKS: 100**

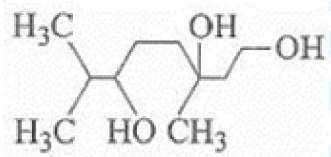
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

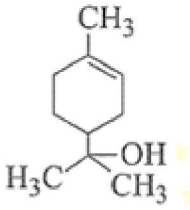
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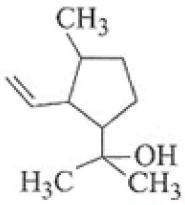
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

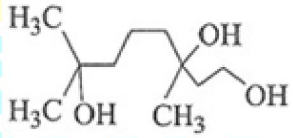
26. The major product of the following reaction is :



A) 

B) 

C) 

D) 

27. In chromatography, which of the following statement is incorrect for Rf?

A) Rf value depends on the type of chromatography

B) The value of R_f cannot be more than one

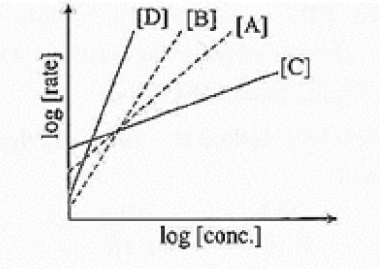
C) Higher Rf value means higher adsorption

D) Rf value is dependent on the mobile phase

28. Consider the following reaction $A \rightarrow P_1$, $B \rightarrow P_2$, $C \rightarrow P_3$, $D \rightarrow P_4$, The order of the above reactions are a, b, c and d respectively, The following graph is obtained when log[rate] Vs log[conc.] are plotted

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Among the following, the correct sequence for the order of the reactions is:

A) $d > b > a > c$ B) $a > b > c > d$ C) $c > a > b > d$ D) $d > a > b > c$

29. Europium (Eu) and Terbium (Tb) attain state $4f^7$ configuration by exhibiting oxidation states of

A) +2 and +4 B) +3 and +4 C) +2 and +3 D) +1 and +3

30. Choose the correct statement

A) CCl_4 is non-polar because C-Cl bond is non-polar

B) CCl_4 is polar because C–Cl bond is polar

C) Though C–Cl bond is polar, CCl_4 is non polar

D) Though C–Cl bond is non-polar, CCl_4 is polar

31. A buffer solution is prepared by mixing 10 ml of 1 M CH_3COOH and 20 ml of 0.5 M CH_3COONa and then diluted to100ml with distilled water. If the pK_a of CH_3COOH is 4.76, what is the pH of the buffer solution prepared?

A) 3.76 B) 4.76 C) 5.8 D) 6.4

32. Highest oxidation state of Mn is exhibited in Mn_2O_7. The correct statement about Mn_2O_7 is

A) Mn is tetrahedrally surrounded by oxygen atoms and vice versa

B) Mn is octahedrally surrounded by oxygen atoms

C) Contains Mn– O – Mn bridge with larger bond length of (Mn – O) bond

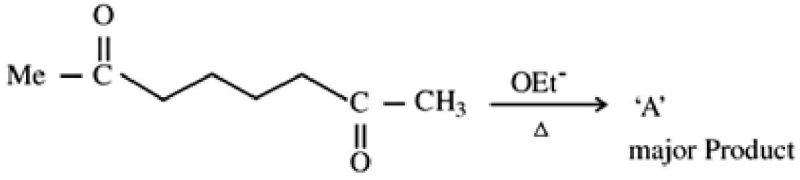
D) Contains Mn– O –Mn bond angle 109°28'

33. Which of the following complex will show largest splitting of d-orbitals?

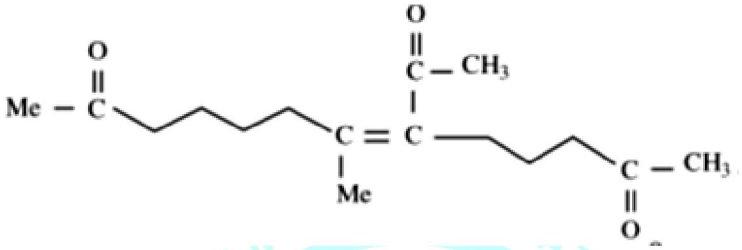
A) $[FeF_6]^{3-}$ B) $[Fe(NH_3)_6]^{3+}$ C) $[Fe(CN)_6]^{3-}$ D) $\left[Fe(C_2O_4)_3\right]^{3-}$

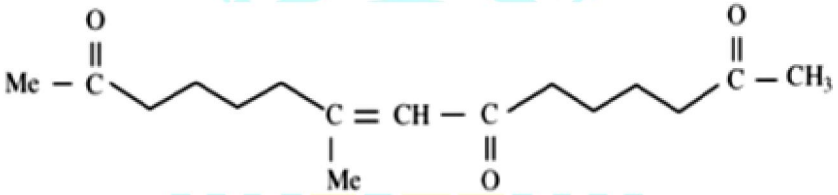
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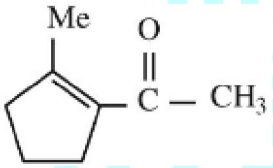
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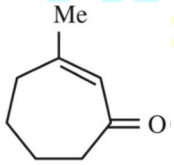
34. 

A in the above reaction is :

A) 

B) 

C) 

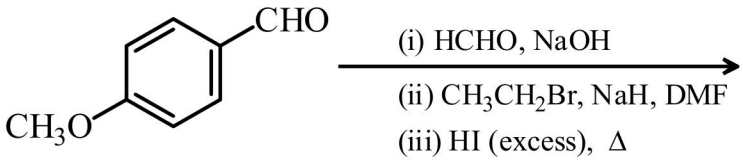
D) 

35. The IUPAC name of ethylidene chloride is:

A) 1-Chloroethene B) 1-Chloroethyne

C) 1,2-Dichloroethane D) 1,1-Dichloroethane

36. Identify A in the given chemical reaction

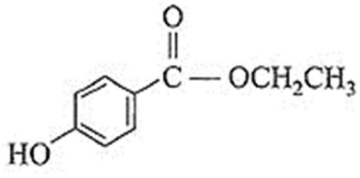


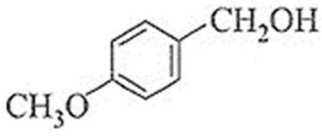
A

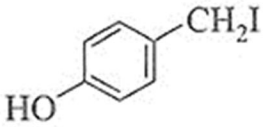
(It contains Benzene ring)

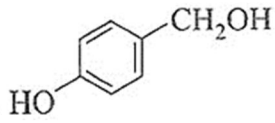
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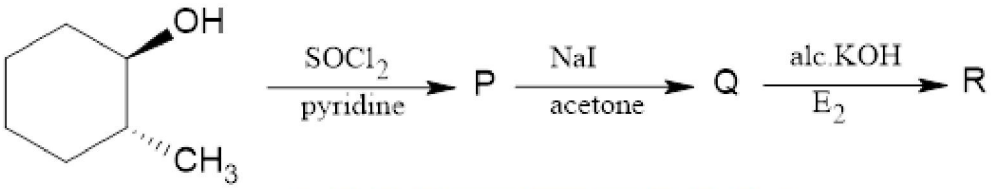
A) 

B) 

C) 

D) 

37. The **incorrect** statement about the following sequence of reactions is?



A) The product P is formed by retention of configuration

B) The product Q is formed by inversion of configuration

C) The major product R is a single optically active compound

D) The major product R is formed by Hoffmann rule

38. Match List-I with List-II and select the correct answer using the codes given below the lists (l and m are respectively the azimuthal and magnetic quantum no.)

List-I		**List-II**	
(P)	Number of values of l for a given energy level	1.	0, 1, 2, ….. (n – 1)
(Q)	Value of l for a particular type of orbit	2.	$-l$ to $+l$ through zero
(R)	Number of values of m for $l = 2$	3.	5
(S)	Value of 'm' for a particular type of subshell	4.	n

A) $P-4; Q-1; R-2; S-3$

B) $P-4; Q-1; R-3; S-2$

C) $P-1; Q-4; R-2; S-3$

D) $P-1; Q-4; R-3; S-2$

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39. Two solid compounds X and Y dissociate at a certain temperature as follows

$X(s) \rightleftharpoons A(g) + 2B(g);\ K_{p_1} = 9\times10^{-3}\ atm^3$

$Y(s) \rightleftharpoons 2B(g) + C(g);\ K_{p_2} = 4.5\times10^{-3}\ atm^3$

The total pressure of gases over a mixture of X and Y is :

A) 4.5 atm B) 0.45 atm C) 0.6 atm D) None of these

40. Na-amalgam is prepared by electrolysis of NaCl solution using liquid Hg as cathode. How long should the current of 10 amp (100% efficiency) is passed to produce 10% Na-Hg on a cathode of 10 gm Hg. (atomic mass of Na = 23)

A) 7.77 min B) 9.44 min C) 5.24 min D) 11.39 min

41. Calculate the value of $\Lambda_m^\infty\ \left(\Omega^{-1}cm^2mol^{-1}\right)$ for $SrCl_2$ (strong electrolyte) in water at 25°C from the following data :

Conc. (mol/lt)	0.25	1
$\Lambda_m\left(\Omega^{-1}cm^2mol^{-1}\right)$	260	250

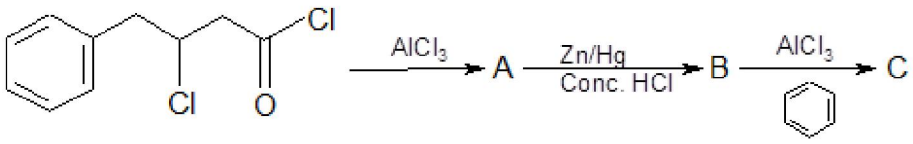
A) 270 B) 260 C) 250 D) 255

42. Which of the following order is not correct?

A) IE(I) of Be > IE(I) of B B) IE(II) of Be > IE(II) of B

C) IE(II) of O > IE(II) of N D) IE(I) of Mg > IE(I) of Al

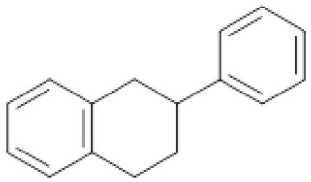
43.

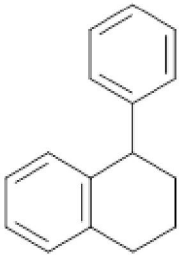


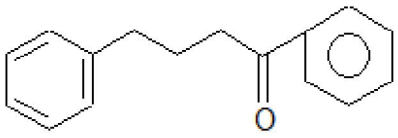
The major product C is :

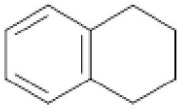
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A) 

B) 

C) 

D) 

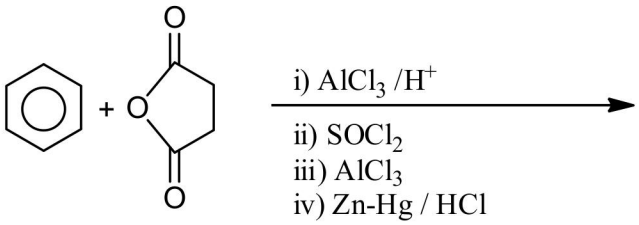
44. An amino acid is characterized by two pKa values the one corresponding to the more acidic site is designated as pKa_1 and the other corresponding to the less acidic site is designated as pKa_2.

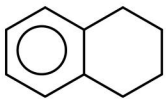
Some amino acids have side chain with acidic or basic groups. These amino acids have pKa_3 value also for the side chain.

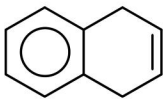
S.No	Amino acid	p^{Ka_1}	p^{Ka_2}	p^{Ka_3} (side chain)
I	Aspartic acid	1.88	9.6	3.65
II	Glutamic acid	2.19	9.67	4.25
III	Lysine	2.18	8.95	10.53
IV	Arginine	2.17	9.04	12.48

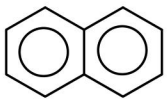
The isoelectric point (pI) of Aspartic acid and lysine will be respectively :

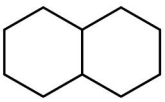
A) 6.62 & 9.74 B) 2.77 & 5.6 C) 2.77 & 9.74 D) 9.74 & 6.62

45. 

A) 

B) 

C) 

D) 

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. The volume, in mL, of 0.02 M $K_2Cr_2O_7$ solution required to oxidized completely react with 0.288 g of ferrous oxalate in acidic medium is _____.
(Molar mass of Fe = 56, O=16, C=12 g mol^{-1})

47. For silver, $C_p\left(JK^{-1}mol^{-1}\right) = 23 + 0.01T$. If the temperature (T) of 3 moles of silver is raised from 300 K to 1000 K at 1 atm pressure, the value of ΔH will be ….kJ/mol? (Nearest integer)

48. Find the number of allotropes which contain discrete molecules held together by London forces:
Diamond, Graphite, Fullerenes, White phosphorus, Rhombic Sulphur, Black Phosphorus.

49. Number of electrons present in Thorium atom with $l = 3$ is

50. How many isomers are possible for the complex, $\left[Pt(NH_3)_2(SCN)_2\right]$?

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SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. If the circle $x^2+y^2-2gx+6y-19c=0,$ passes through the point (6, 1) and its centre lies on the line x – 2cy = 8, then the length of intercept made by the circle on x-axis is

A) $\sqrt{11}$ B) 4 C) 3 D) $2\sqrt{23}$

52. The value of $\lim\limits_{n\to\infty}\dfrac{1+2-3+4+5-6+.....+(3n-2)+(3n-1)-3n}{\sqrt{2n^4+4n+3}-\sqrt{n^4+5n+4}}$

A) $\dfrac{\sqrt{2}+1}{2}$ B) $3\left(\sqrt{2}+1\right)$ C) $\dfrac{3}{2}\left(\sqrt{2}+1\right)$ D) $\dfrac{3}{2\sqrt{2}}$

53. The value of $\cot\left(\sum\limits_{n=1}^{50}\tan^{-1}\left(\dfrac{1}{1+n+n^2}\right)\right)$ is

A) $\dfrac{26}{25}$ B) $\dfrac{25}{26}$ C) $\dfrac{50}{51}$ D) $\dfrac{52}{51}$

54. Let α *and* β be real numbers. Consider a 3×3 matrix A such that $A^2=3A+\alpha I$. If $A^4=21A+\beta I$, then

A) $\alpha=1$ B) $\alpha=4$ C) $\beta=8$ D) $\beta=-8$

55. If the system of equations $2x+y-z=5$

$$2x-5y+\lambda z=\mu$$

$$x+2y-5z=7$$

has infinitely many solutions, then $(\lambda+\mu)^2+(\lambda-\mu)^2$ is equal to

A) 916 B) 912 C) 920 D) 904

56. Let $f:R\to R$ be a function defined by $f(x)=\begin{cases}\lim\limits_{n\to\infty}\sqrt[n]{2^{nx}+x^{2n}} & ; \quad x\geq 0\\ f(-x) & ; \quad x<0\end{cases}$ then the number of non-differentiable points of $f(x)$ is

A) 3 B) 4 C) 5 D) none of these

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57. In ΔABC, $c^2 > a^2 + b^2$ and $\sin^5 A + \cos^5 A > \sin^5 B + \cos^5 B$ then $\left[\frac{a}{b}\right] + \left[\frac{b}{c}\right]$ is (where $[\bullet]$ denotes GIF)

A) 0 B) 1 C) 2 D) can't be determined

58. For $I(x) = \int \frac{\sec^2 x - 2022}{\sin^{2022} x} dx$, *if* $I\left(\frac{\pi}{4}\right) = 2^{1011}$, then

A) $3^{1010} I\left(\frac{\pi}{3}\right) - I\left(\frac{\pi}{6}\right) = 0$ B) $3^{1010} I\left(\frac{\pi}{6}\right) - I\left(\frac{\pi}{3}\right) = 0$

C) $3^{1011} I\left(\frac{\pi}{3}\right) - I\left(\frac{\pi}{6}\right) = 0$ D) $3^{1011} I\left(\frac{\pi}{6}\right) - I\left(\frac{\pi}{3}\right) = 0$

59. Let f(x) be a function satisfying $f(x) + f(\pi - x) = \pi^2, \forall\ x \in R$. Then $\int_0^{\pi} f(x) \sin x dx$ is equal to

A) $\frac{\pi^2}{4}$ B) $\frac{\pi^2}{2}$ C) $2\pi^2$ D) π^2

60. Positive integers a,b and c are randomly and independently selected with replacement from set {1,2,3…..,2010} . What is the probability that abc+ab+a is divisible by 3?

A) $\frac{1}{3}$ B) $\frac{29}{81}$ C) $\frac{11}{27}$ D) $\frac{13}{27}$

61. If position vectors of points P,Q,R are $\bar{i} + 4\bar{j} - 2\bar{k}, 2\bar{i} + \bar{j} - 2\bar{k}, -5\bar{j} + \bar{k}$ respectively then the perpendicular distance from P to QR is_____

A) $\frac{3\sqrt{26}}{7}$ B) $\frac{6\sqrt{23}}{7}$ C) $\frac{1}{3}$ D) 0

62. Let $f(x) = \frac{x.e^x}{e^x - 1}$. If $f(x)$ can be written as sum of an even function, $g(x)$ and an odd function, $h(x)$ then the value of $h(4)$ is

A) 2 B) 1 C) $\frac{4}{\sqrt{e^4 - 1}}$ D) $\frac{4(e^4 + 1)}{e^4 - 1}$

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63. If α *and* β are the roots of the equation, $7x^2 - 3x - 2 = 0$, then the value of $\frac{\alpha}{1-\alpha^2} + \frac{\beta}{1-\beta^2}$ is equal to

A) $\frac{27}{32}$ B) $\frac{1}{24}$ C) $\frac{3}{8}$ D) $\frac{27}{16}$

64. Fractional part of the number $\frac{4^{2022}}{15}$ is equal to

A) $\frac{4}{15}$ B) $\frac{1}{15}$ C) $\frac{14}{15}$ D) $\frac{8}{15}$

65. The 4th term of GP is 500 and its common ratio is $\frac{1}{m}, m \in N$. Let S_n denote the sum of the first n terms of this GP. If $S_6 > S_5 + 1$ and $S_7 < S_6 + \frac{1}{2}$, then the number of possible values of m is ____

A) 12 B) 14 C) 10 D) 16

66. A light ray emits from the origin making an angle 30° with the positive x-axis. After getting reflected by the line x + y = 1, if this ray intersects x-axis at Q, then the abscissa of Q is

A) $\frac{2}{\left(\sqrt{3}-1\right)}$ B) $\frac{2}{\left(3+\sqrt{3}\right)}$ C) $\frac{2}{\left(3-\sqrt{3}\right)}$ D) $\frac{\sqrt{3}}{2\left(\sqrt{3}+1\right)}$

67. Let the eccentricity of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$, *be* $\frac{1}{4}$. If this ellipse passes through the point $\left(-4\sqrt{\frac{2}{5}}, 3\right)$, $a^2 + b^2$ is equal to :

A) 29 B) 31 C) 32 D) 34

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68. Let R be a relation on $N \times N$ defined (a, b) R (c, d) if and only if ad (b – c) = bc (a – d). Then R is

A) symmetric but neither reflexive nor transitive.

B) trasitive but neither reflexive nor symmetric.

C) reflexive and symmetric but not transitive.

D) symmetric and transitive but not reflexive.

69. Let A = {1, 2, 3, ….10} and $f : A \to A$ be defined as

$$f(k) = \begin{cases} k+1 & if\ k\ is\ odd \\ k & if\ k\ is\ even \end{cases}$$

then the number of possible functions $g : A \to A$ such that gof = f is :

A) 5^5 B) 10^5 C) 5! D) ${}^{10}C_5$

70. Let $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$. If |adj (adj (adj 2A))| = $(16)^n$, then n is equal to

A) 10 B) 9 C) 12 D) 8

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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71. The mean and standard deviation of the marks of 10 students were found to be 50 and 12 respectively. Later, it was observed that two marks 20 and 25 were wrongly read as 45 and 50 respectively. Then the correct variance is

72. Let the area enclosed by the x-axis, and the tangent and normal drawn to the curve $4x^3 – 3xy^2 + 6x^2 – 5xy – 8y^2 + 9x + 14 = 0$ at the point (-2, 3) be A. Then 8A is equal to

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73. Let y = y(x) be a solution of the differential equation

(x cos x)dy + (xy sin x + y cos x – 1) dx = 0, $0 < x < \frac{\pi}{2}$. If $\frac{\pi}{3} y\left(\frac{\pi}{3}\right) = \sqrt{3}$, then

$\left|\frac{\pi}{6} y''\left(\frac{\pi}{6}\right) + 2y'\left(\frac{\pi}{6}\right)\right|$ is equal to

74. If $z^2 + z + 1 = 0, z \in C$, *then* $\left|\sum_{n=1}^{15}\left(z^n + (-1)^n \frac{1}{z^n}\right)^2\right|$ is equal to

75. The number of ways, 16 identical cubes, of which 11 are blue and rest are red, can be placed in a row so that between any two red cubes there should be at least 2 blue cubes, is

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